

Digital Payment Transaction Analysis using SQL

Subtitle: Transforming Paypal Transaction Data into Business Insights for Strategic Decision-Making

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Role: Data Analyst

Tools Used

- MySQL
- SQL
- GitHub
- Gamma(presentation)
- Google docs(Report making)

Skills Demonstrated

Technical Skills

- SQL
- Joins
- CTEs
- Aggregate Functions
- CASE Statements
- GROUP BY
- HAVING
- Date Functions
- Data Cleaning
- Data Modeling

Business Skills

- Business Analysis
- Critical Thinking
- Problem Solving
- Data Storytelling
- Data Interpretation
- Business Intelligence
- Strategic Recommendations
- Business Analytics
- Data Interpretation

- Business Intelligence

Project Overview

Objective: The goal of this project is to analyze PayPal transaction data using SQL to uncover actionable insights that support strategic decision-making, improve risk management, optimize merchant and customer performance, and ultimately increase the platform's market share and improve platform performance.

Business Goals

- Identify key geographic markets
- Evaluate merchant performance
- Analyze customer behavior
- Monitor transaction trends
- Identify valuable customers
- Analyze currency exposure
- Improve customer engagement
- Support strategic business decisions

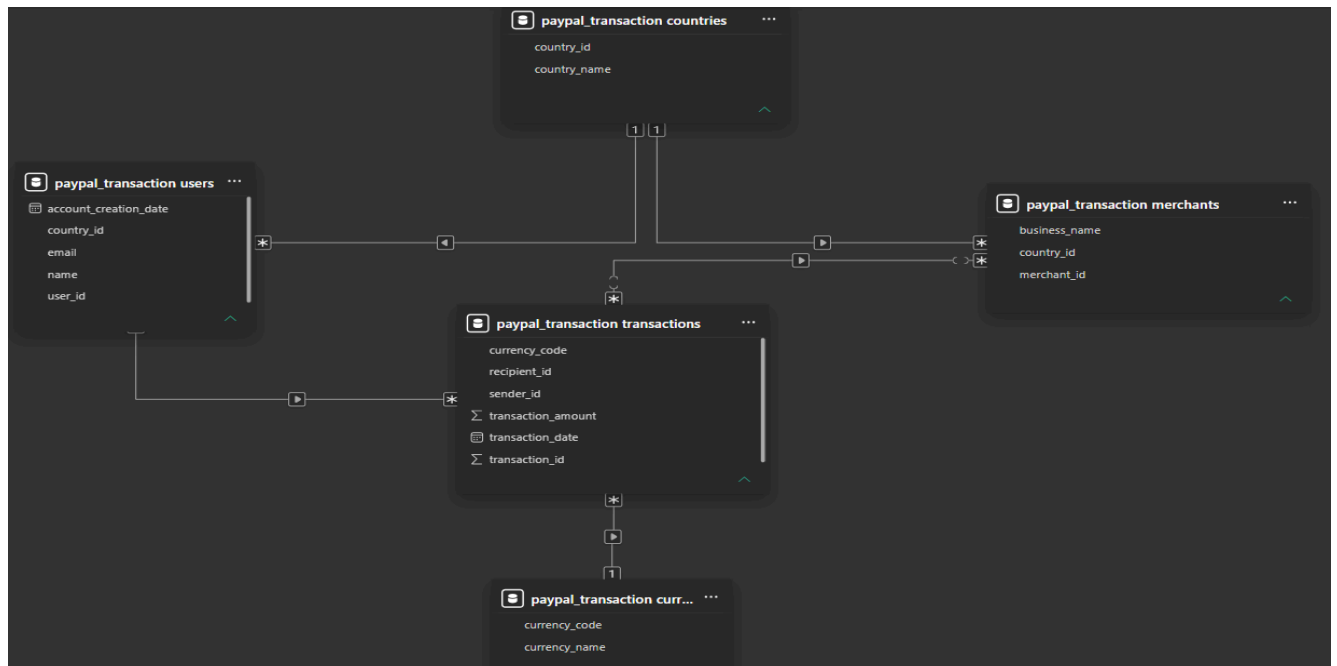
Dataset Overview

Database Tables

Dataset: [paypal transaction link](#)

Table	Description
Users	Customer information
Transactions	Payment transactions
Merchants	Merchant details
Countries	Country information
Currencies	Currency details

Database Schema (ER Diagram)



Business Questions

1. Which countries generated the highest transaction value?

country_name	totaltransaction
Zambia	174263.15
Iceland	140067.63
Saint Kitts and Nevis	121171.01
Timor-Leste	118107.68
Guadeloupe	98715.18

Transactions came from several countries, with Zambia, Iceland, Saint Kitts and Nevis, Timor-Leste, and Guadeloupe contributing the highest transaction value.

2. Which currencies handled the highest transaction volume?

Result Grid			Filter Rows:	
	currency_code	total_converted		
▶	EUR	2583686.81		
	JPY	2326288.59		
	USD	2251330.4		

EUR had the highest converted transaction amount, followed by JPY and USD.

3. Which merchants received the highest payments?

Result Grid					Filter Rows:		Export:	Wrap Cell Content:
	merchant_id	business_name	total_recieved	average_transaction				
▶	151	Gould LLC	36380.25	9095.06				
	91	Wilson-Mosley	32675.13	6535.03				
	129	Grant-Gallegos	29428.43	4904.74				
	24	Simon PLC	28920.03	5784.01				
	53	Shelton, Jones and Ferguson	28176.04	7044.01				
	17	Bender Ltd	26454.78	8818.26				
	25	Cherry and Sons	24795.12	4959.02				
	113	Melton, Johnston and Lee	23079.2	5769.8				
	102	Knapp, Dean and Jones	21902.53	4380.51				
	98	Lopez-Green	21896.18	5474.04				

A few merchants generated the highest revenue. Gould LLC received the highest total transaction amount among all merchants.

4. Which merchants exceeded the performance threshold?

Result Grid						Filter Rows:		Export:	Wrap Cell Content:
	merchant_id	business_name	transaction_year	transaction_month	total_transaction_amount	performance_status			
▶	1	Hunter, Orr and Serrano	2024	1	5651.58	Did not exceed \$50000			
	1	Hunter, Orr and Serrano	2024	4	9517.07	Did not exceed \$50000			
	2	Bauer Ltd	2023	11	2920.89	Did not exceed \$50000			
	2	Bauer Ltd	2023	12	1160.22	Did not exceed \$50000			
	2	Bauer Ltd	2024	2	871.17	Did not exceed \$50000			
	2	Bauer Ltd	2024	3	2672.22	Did not exceed \$50000			
	3	Gates, Hill and Murphy	2023	12	1074.97	Did not exceed \$50000			
	4	Walker-Lynn	2024	1	4074.83	Did not exceed \$50000			
	7	Castro, Green and Jones	2023	11	10417.53	Did not exceed \$50000			
	7	Castro, Green and Jones	2024	1	7512.52	Did not exceed \$50000			
	8	Harris-Nguyen	2024	1	6269.37	Did not exceed \$50000			
	8	Harris-Nguyen	2024	4	7400.6	Did not exceed \$50000			
	9	Baker Inc	2023	11	15130.89	Did not exceed \$50000			

No merchant exceeded the predefined monthly transaction threshold of \$50,000 during the analyzed period, indicating that merchant transaction volumes remained below the target benchmark.

5. How do monthly transaction amounts change over time in 2023?

Result Grid			
Filter Rows:			
Export:			
Wrap			
	transaction_year	transaction_month	total_amount
▶	2023	1	963877.18
	2023	2	955585.82
	2023	3	962664.39
	2023	4	919967.51
	2023	5	974664.83
	2023	6	806503.91
	2023	7	1037697.76
	2023	8	914028.09
	2023	9	885073.79
	2023	10	1100563.77
	2023	11	1027723.67
	2023	12	835610.04

The platform maintains consistent monthly transaction activity, with monthly transaction values generally ranging between \$80,000 and \$110,000, suggesting a stable customer base and steady transaction flow throughout the year.

6. Who are the highest spending customers?

Result Grid	
Filter Rows:	
Export:	
Wrap Cell Content:	
user_id	email
▶	51 andrewgonzales@gmail.com
	133 xarnold@kim.com
	179 allisonmiller@fernandez.org
	283 cmcdonald@jones.net
	303 wcampbell@yahoo.com
	350 brittanyturner@gmail.com
	520 smithjoseph@hotmail.com
	523 taylorchristopher@perkins.info
	686 john30@clark-norris.biz
	701 jenna87@hart.com
	718 steve24@yahoo.com
	819 richardsoncraig@hernandez.com
	892 melissahughes@white.com

7. What is the ratio of domestic and international transactions?

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130 FROM users u1
139 JOIN transactions t

```

transaction_type	transaction_count
International	587
Domestic	9

Most transactions on the platform are international rather than domestic, showing that the platform is mainly used for cross-border payments.

8. How many high-value transactions occurred?

transaction_category	transaction_count
Regular International	2233
Regular Domestic	45
High Value International	1

Regular International transactions (2,233) account for the vast majority of payment volume, while Regular Domestic transactions (45) and High-Value International transactions (1) represent only a small fraction of the total transaction portfolio.

9. Which customers are highly engaged?

user_id	email	name	total_amount
970	brittanywright@morales.net	Dwayne Stephenson	232577.58

Key Findings

- Most transactions on the platform are **international** rather than domestic, showing that the platform is mainly used for cross-border payments.

- **Regular international transactions** make up most of the transaction volume, while high-value transactions are very few.
- A few merchants generated the highest revenue. **Gould LLC** received the highest total transaction amount among all merchants.
- Monthly transaction amounts remained fairly stable, with no major increase or decrease throughout the period.
- Transactions came from several countries, with **Zambia, Iceland, Saint Kitts and Nevis, Timor-Leste, and Guadeloupe** contributing the highest transaction values.
- **EUR** had the highest converted transaction amount, followed by **JPY** and **USD**.
- A small number of customers had much higher transaction amounts than other users, making them valuable customers for the platform.

Business Insights

- The platform is mainly used for **international payments**, so improving international payment services could help the business grow.
- Since only a few merchants generate most of the revenue, helping other merchants increase their sales could improve overall platform performance.
- Since **EUR transactions are the highest**, Europe appears to be an important market for the platform.
- High-value transactions are rare but should still be monitored carefully because they involve larger amounts of money.
- High-spending customers are important to the business and could be targeted with loyalty programs or special offers.

Project Summary

This project demonstrates how SQL can transform raw transaction data into meaningful business intelligence. By analyzing customer behavior, merchant performance, transaction trends and geographic markets, the analysis provides actionable recommendations that support data-driven decision-making and supports business growth.

Thank you!

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